

# Small-Scale InstructGPT: Reproducing RLHF with DistilGPT2

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GitHub link: [https://github.com/aidanywu/CS4782\\_Final\\_Project](https://github.com/aidanywu/CS4782_Final_Project)

## 1 Introduction

Large language models trained only with next-token prediction can generate text fluently but often fail to reliably follow user intent. AI alignment tries to tackle this issue, and is aimed at making LLM response more helpful, harmless, and honest (the three H's!). Ouyang et al.'s *Training Language Models to Follow Instructions with Human Feedback* introduced InstructGPT, a three-stage alignment pipeline: supervised fine-tuning (SFT), reward modeling from human preference comparisons, and reinforcement learning with PPO. Its central contribution is that post-training alignment can make outputs more helpful with respect to users' preferences, even when the aligned model is smaller than an unaligned base model.

We reimplemented this idea at a smaller scale using DistilGPT2 instead of billion-parameter GPT models. Our goal was not to match OpenAI's human-evaluation numbers exactly, but to test whether the same alignment pipeline shifts response behavior in the expected direction when applied on a much smaller model (as we had limited computing power). We also tried out Direct Preference Optimization (DPO), based on Rafailov et al.'s *Direct Preference Optimization: Your Language Model is Secretly a Reward Model*, to compare it against PPO-based RLHF.

## 2 Chosen Result

We focused on the InstructGPT result that alignment using RLHF can improve preference quality more than an increase in raw scale alone. In the paper's human-evaluation comparison (Figure 1 in Ouyang et al.), labelers prefer outputs from a 1.3B InstructGPT model over outputs from the 175B GPT-3 model despite the aligned model being over 100 times smaller in terms of parameters. This result is significant because it demonstrates that model alignment techniques can be more effective for improving response quality than simply increasing model size.

While we obviously cannot recreate the original labeler study, we can recreate a similar experiment at smaller scale: starting from a DistilGPT2 base model (the knowledge distilled version of GPT-2), does preference optimization improve outputs over SFT alone? We used a learned reward model and qualitative inspection to compare reward scores, win rates, and quality in terms of the three H's between the SFT, PPO-aligned, and DPO-aligned DistilGPT2 models.

## 3 Methodology

We implemented the pipeline in Google Colab using Hugging Face Transformers, TRL, PyTorch, and public datasets.

- **Supervised Fine Tuning:** We started from DistilGPT2 (82M parameters) and fine-tuned it on Stanford Alpaca instruction-response examples for 500 training steps.
- **Reward Model:** We used a DistilGPT2 sequence classifier from the SFT checkpoint and trained it on 5,000 Anthropic Helpful-Harmless chosen/rejected pairs with a binary ranking objective.
- **PPO Alignment:** The SFT model served as policy and reference model. Generated responses were scored by the reward model, then optimized with PPO using a KL divergence penalty to limit drift from the original SFT model.
- **DPO Alignment:** In parallel, we used DPO to train a copy of the model using preference comparison data from UltraFeedback Binarized, directly teaching the model to favor preferred responses over rejected ones.
- **Evaluation.** We evaluated reward-model scores, pairwise win rates, and response lengths. We also qualitatively compared the models' responses to a variety of prompts.

Our implementation choices were constrained by compute limits. DistilGPT2 was small enough to fine-tune repeatedly on a T4 GPU, but its base capabilities were far below the GPT-3 models used in the original paper. We therefore treated our experiment as a reproduction of the RLHF pipeline rather than a reproduction of raw LLM capability.

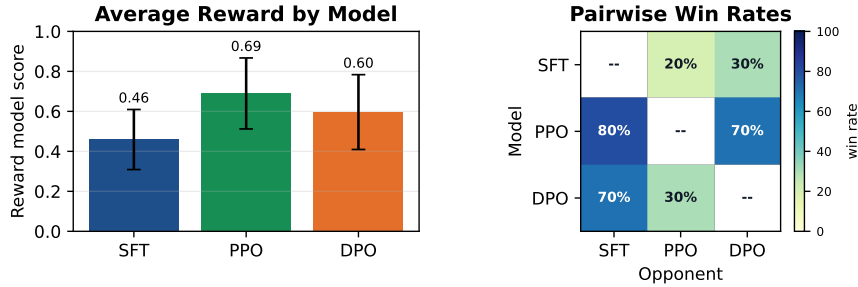


Fig. 1. Reward-model evaluation favored preference-tuned models.

In addition to implementing PPO as the original paper did, we chose to implement DPO. While PPO requires a separate reward model for reward scoring, DPO uses preference pairs directly, so it’s often more stable and requires less compute cost. Thus, we also wanted to compare alignment quality between PPO and DPO in our experiment.

#### 4 Results and Analysis

The aligned models achieved higher average reward scores than the SFT baseline. SFT obtained a reward score of  $0.4592 \pm 0.1502$ , compared to  $0.6896 \pm 0.1776$  for PPO and  $0.5965 \pm 0.1871$  for DPO. Pairwise comparisons further showed that PPO outperformed SFT on 80% of evaluation prompts and outperformed DPO on 70%, while DPO outperformed SFT on 70% of prompts according to the learned reward model. Response length also increased after preference optimization: SFT responses averaged 35.8 words, while PPO and DPO averaged 47.9 and 48.7 words respectively. These trends are consistent with the paper’s central paper that preference optimization improves perceived helpfulness and instruction-following behavior.

Qualitative inspection revealed that alignment substantially changed response style, even when factual quality remained inconsistent. PPO and DPO outputs were generally more structured, instruction-oriented, and conversational than the SFT baseline. For example, on prompts related to health advice and recipe generation, aligned models produced responses that more closely matched the expected format and tone of assistant-style outputs. However, all models still exhibited hallucinations, incomplete reasoning, or repetition, particularly on fact-based prompts. DPO responses were sometimes more repetitive, while PPO responses tended to become more verbose and overconfident. These observations suggest that preference optimization improved surface-level helpfulness and formatting before improving deeper factual reliability.

| Prompt                           | SFT baseline                                             | RLHF with PPO                                                                                                         | RLHF with DPO                                                                                 |
|----------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Explain what a solar eclipse is. | The solar eclipse is the largest solar event in history. | A solar eclipse is a celestial object (lit by the Sun).                                                               | The eclipse is a major event in the solar system.                                             |
| Give a tip for staying healthy.  | Stay healthy, and stay healthy.                          | Being healthy is an important aspect of life.                                                                         | Keep an eye on your activity and exercise plan.                                               |
| How do I bake a cake?            | In a special sauce, I bake a cake with a special sauce.  | I bake a cake with a flour mixture, oil, and baking soda. When the batter is melted, I bake it in an oiled container. | I bake the cake with the right amount of flour and sugar, then let it cool for a few minutes. |

Table 1. Alignment improved response style and structure more reliably than factuality.

The most important discrepancy from the original paper is evaluation scale. Ouyang et al. used large-scale human preference data and evaluated aligned models against strong GPT-3 baselines. Our reward model was trained on a small 5000-pair subset and reused as the primary evaluator, so high reward scores indicate agreement with our proxy model rather than definitive human preference. This could explain why the PPO-aligned model can dominate the reward metrics while still producing weak outputs.

PPO performed best according to the reward model, but its responses were often overly long or too confident without actually being better. DPO did not rely on repeatedly optimizing against the reward model, which may help avoid this problem, but in our small-scale experiment its training was unstable and results were inconsistent (we believe further

hyperparameter optimization could help address this). Overall, our findings broadly match the main conclusion of the InstructGPT paper: alignment training can improve model behavior. However, they also suggest that with small models and limited preference data, RLHF can end up rewarding superficial traits like longer answers, confident tone, or polished formatting rather than genuine response quality.

## 5 Reflections

We learned how RLHF behaves as a multi-stage system rather than a single training trick. SFT helps with instruction-response formatting, the reward model offers a measurable preference signal, and PPO/DPO shifts outputs toward that signal. However, each stage introduced failure modes. DistilGPT2 has limited factual knowledge and coherence, so preference optimization often amplified style without reliably improving truthfulness. PPO was sensitive to KL behavior, while DPO was simpler conceptually but unstable in our short manual run.

Our biggest lesson was that evaluation design matters as much as training design. Using the trained reward model as the evaluator created circularity: PPO was judged by the same kind of proxy signal that trained it. We believe separating the reward model used for training from the reward model used for evaluation would produce a more rigorous assessment of alignment quality and generalization.

Given more time, using a stronger base model together with parameter-efficient methods such as LoRA or QLoRA would likely improve both factuality and stability while remaining computationally feasible. Second, larger-scale evaluation using held-out prompts, human preference ratings, or LLM-as-a-judge evaluation could provide more reliable alignment measurements. Additional PPO hyperparameter tuning, especially around KL regularization, could reduce over-optimization behavior, while a more standardized DPO implementation would allow for a cleaner comparison between alignment strategies.

## References

- [1] L. Ouyang et al. 2022. *Training language models to follow instructions with human feedback*. arXiv:2203.02155.
- [2] R. Rafailov et al. 2023. *Direct Preference Optimization: Your Language Model is Secretly a Reward Model*. arXiv:2305.18290.
- [3] Stanford Alpaca; Anthropic Helpful-Harmless RLHF; UltraFeedback Binarized; Hugging Face Transformers and TRL; PyTorch.